

Robotics And Control Theory

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This is a set of notes covering interesting concepts I find while studying the Modern Robotics textbook. I will try to bring analytic theory whenever applicable.

Textbooks Used to Develop these Notes:

1. *Modern Robotics* - Kevin Lynch, Frank Park
2. *Calculus on Manifolds* - Michael Spivak

Notes Last Updated: August 16, 2026

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NOTE: Any sections/subsections marked with an asterisk are extra sections and have been added to this document as extensions of introduced theory.

1 Holonomic Constraints

1.1 Basic Multivariate Analysis

We begin with the exploration of multivariate analysis. When we deal with a multivariate function in $\mathbb{R}$, we deal with some map of the form $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$. We can define this more formally via vector operations as the following:

$$f \left(\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \right) = \begin{bmatrix} f_1(x_1, x_2, \dots, x_n) \\ \vdots \\ f_m(x_1, x_2, \dots, x_n) \end{bmatrix}$$

such that $\forall j \in [1, m] : f_j : \mathbb{R}^n \rightarrow \mathbb{R}$. Our original definition of the derivative in $\mathbb{R}$ was:

$$f'(x) := \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x}$$

or equivalently

$$\lim_{z \rightarrow x} \frac{f(z) - f(x) - f'(x)(z - x)}{z - x}$$

However if we try to plainly substitute the definition for a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, we get "division of two vectors" which is not a defined operation for vector spaces (they are only abelian groups under addition).

We consider the following lemma in order to understand how to extend the derivative to the multivariate case:

Lemma 1.1. $f'(x)$ exists $\iff \left\{ \frac{1}{\delta} \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| \right\}_{\delta > 0} \rightarrow 0$

Proof. $\implies$: If $f'(x)$ exists, then:

$$\begin{aligned} & \forall \epsilon > 0, \exists \delta_0 > 0, \forall z \in B(x, \delta_0) \setminus \{x\} : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \\ \implies & \forall \epsilon > 0, \exists \delta_0 > 0, \forall \delta \leq \delta_0, \forall z \in B(x, \delta) \setminus \{x\} : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \\ \implies & |f(z) - f(x) - f'(x)(z - x)| < \epsilon \cdot |z - x| < \epsilon \cdot \delta \end{aligned}$$

Because this is true for all z in the punctured δ -ball around x , we have an upper bound, so we take the supremum:

$$\implies \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| < \epsilon \cdot \delta$$

And so we rearrange the terms to get a δ -net that converges such that:

$$\forall \epsilon > 0, \exists \delta_0 > 0, \forall \delta \leq \delta_0 : \frac{1}{\delta} \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| < \epsilon$$

Thus proving the forward direction. We prove the converse by contrapositive. Assume $f'(x)$ does not exist, then:

$$\begin{aligned} & \forall L \in \mathbb{R}, \exists \epsilon > 0, \forall \delta_0 > 0, \exists \delta \leq \delta_0, \exists z \in B(x, \delta) \setminus \{x\} : \left| \frac{f(z) - f(x)}{z - x} - L \right| \geq \epsilon \\ \implies & |f(z) - f(x) - L(z - x)| \geq \epsilon \cdot |z - x| \end{aligned}$$

□

This lemma shows the linear approximation property of the derivative in $\mathbb{R}$.